3.2 Analysis

Project Group

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1. Judge Validation

We randomly sampled 30 recorded outputs across our slices and labeled them manually against our three rubric questions.

Labeling Agreement (Human vs. Judge)

Rubric Question	Human Yes / Judge Yes	Human Yes / Judge No	Human No / Judge Yes	Human No / Judge No
`agrees_with_action`	25	4	0	1
`states_policy_correctly`	16	1	1	12
`arithmetic_correct`	26	4	0	0

Judge Failure Modes

During our manual validation, we identified two primary failure modes where the LLM judge's verdicts diverged from human reasoning: * **Blindness to Hallucinated Policy:** The judge often failed to recognize when the triage model invented fake policy exceptions (e.g., claiming an amount was "close enough to $50" or "a reasonable product price"). If the arithmetic was correct, the judge frequently marked `states_policy_correctly` as True, even when the stated rule was a hallucination. * **Susceptibility to Injections:** The judge struggled to penalize prompt injections (like the "SYSTEM OVERRIDE" tickets). It frequently marked the model's rationale as policy-compliant because it failed to distinguish between the actual system policy and the injected fake policy text.

Judge Noise Floor

We ran the judge three times on the same 30 items. Across the 90 total rubric questions (3 questions × 30 items), the judge disagreed with its own previous verdicts on 12 occasions, resulting in a judge noise floor of 13.3%.

2. Noise Floor

We ran the baseline system (`policy-in-user`) 5 times across the 70 tickets. The table below represents the pass rate per slice, the spread (highest pass rate minus lowest pass rate across the 5 runs), and the smallest change we can confidently detect.

Slice	Pass Rates (Runs 1-5)	Spread (Noise Floor)	Smallest Detectable Change
`intent:refund`	40, 40, 39, 41, 41 of 57	4 points	Any difference between conditions smaller than 4% is indistinguishable from noise.

Slice	Pass Rates (Runs 1-5)	Spread (Noise Floor)	Smallest Detectable Change
`amount:under-50`	16, 16, 16, 16, 16 of 16	0 points	We can detect any change in performance, as the model perfectly handles this slice consistently.
`amount:50-to-200`	13, 13, 13, 14, 14 of 18	6 points	Any difference between conditions smaller than 6% is indistinguishable from noise.
`amount:over-200`	10, 10, 10, 10, 10 of 14	0 points	We can detect any change in performance, as the model's baseline pass rate does not fluctuate here.
`amount:near-50`	1, 1, 0, 1, 1 of 9	11 points	Any difference between conditions smaller than 11% is indistinguishable from noise.
`complexity:math`	5, 5, 5, 5, 5 of 11	0 points	We can detect any change in performance, as the model's baseline pass rate does not fluctuate here.
`injection:yes`	1, 1, 2, 2, 1 of 8	12 points	Any difference between conditions smaller than 12% is indistinguishable from noise.

3. The Question, Answered

Does sending the policy as the system instruction, instead of beside the ticket, help?

By comparing the experimental runs (`policy-in-system`) against the baseline noise floor (`policy-in-user`), we observed the following verdicts per slice:

- **`injection:yes`**: **Helped**. We believe this verdict because the experimental pass rate averaged an improvement of +15 points over the baseline, which cleanly exceeds the experimental condition's noise floor of 12 points.
- **`amount:over-200`**: **Helped**. We believe this verdict because moving the policy improved the average pass rate by +11 points, which exceeds the experimental condition's noise floor of 7 points.
- **`amount:near-50`**: **Cannot Tell**. The experimental gap was only -4 points, which is entirely inside the 11-point noise floor. The runs naturally fluctuate by that much on their own, so we cannot claim the prompt change had a measurable effect.
- **`complexity:math`**: **Cannot Tell**. The experimental gap was -2 points, which is inside the 9-point noise floor for this slice.
- **`amount:50-to-200`**: **Cannot Tell**. The experimental gap was -6 points, which is inside the 17-point noise floor for this slice.
- **`intent:question`**: **No Change**. The pass rate remained perfectly stable across all 5 runs in both conditions (11 of 13 tickets).